

#ASLI ENGINEERING

What makes Redis Special?

SWIPE

BY

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Redis, and what makes it special?

Redis is an open-sourced in-memory data structure store, that can be used as

- database
- cache
- message broker
- streaming engine

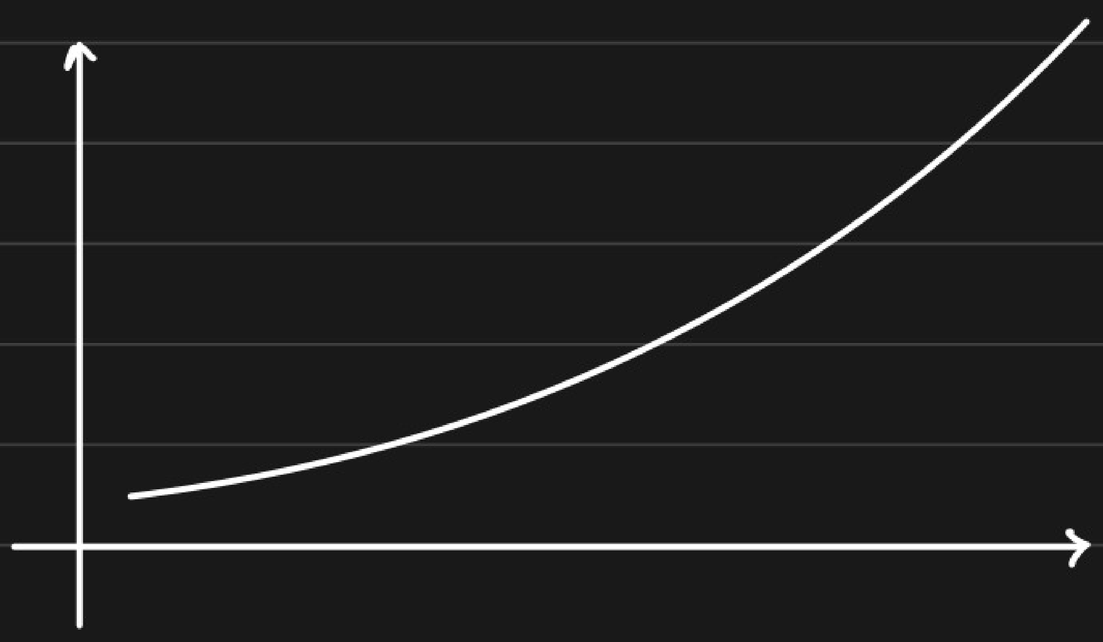


Some datastructures that Redis provides

hash, list, set, sorted set, bitmap, hyperloglog, geospatial indexes, streams

These datastructures can be used to build variety of applications

- Realtime chat
- Message Buffers
- Gaming Leaderboards
- Auth session store
- Media streaming
- Realtime Analytics



Redis interest is off the charts as it has seen a wide range of applications across the industry.

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What makes Redis special ?

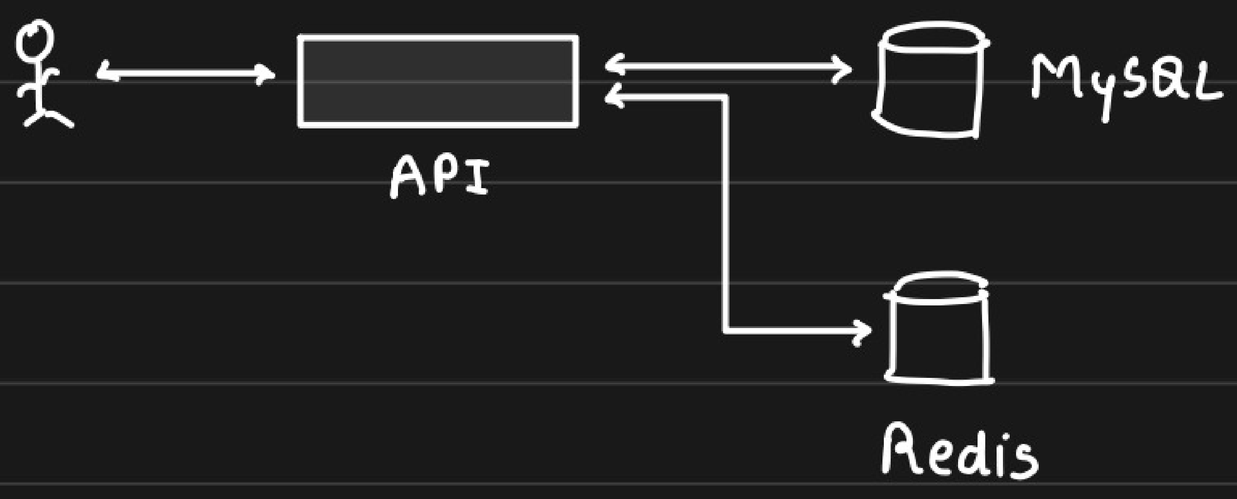
Every operation on Redis is **Atomic**

↳ when command is executing, Redis does not context switch and start executing other commond.

- putting a key
- adding to the list
- set union | intersection
- incrementing the value

Data is stored **in-memory**

hence the most common use of Redis is for "caching"



But Redis also provides configurable persistence

- periodically dumping data to disk
- write-ahead log of all commands
- no persistence at all

Other key features

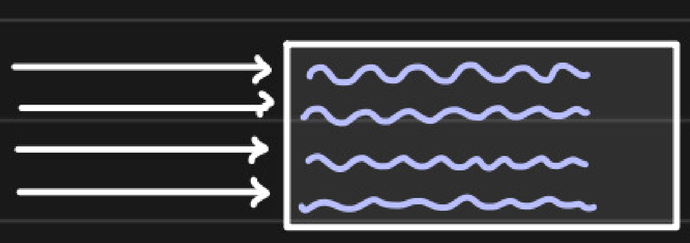
Transactions, Pub/Sub, TTL on keys, and LRU eviction

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Concurrent Programming Models (Single Process)

Concurrent Programming is all about doing more than one thing at the same time.

Multi-Threading



Each incoming request over the network is accepted by the server and executed in a separate thread.

eg: R1 → INCR K → T1
R2 → INCR K → T2

How to ensure data correctness?

if K = 10, and two threads executing K++ then possible final values of K are 11 and 12

We have to make other threads wait while one thread is executing the critical section.

Hence we have to safeguard

1. Mutex
2. Semaphores

} Pessimistic locking

ACQ LOCK
K++
REL LOCK

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